A Position- and Scale-Aware Object-Agent Classifier via Deep Reinforcement Learning for Remote Sensing Imagery

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***Abstract*—Deep object classification is an effective method for land-cover classification in remote sensing images, requiring fewer samples and achieving clearer classification boundaries. However, determining the size and position of the image patches is crucial for accurate deep object classification, and this is often reliant on expert knowledge and trial and error, which limits its general applicability. In this article, a position- and scale-aware object-agent classifier for remote sensing imagery—named ObjectAgents—is proposed. Unlike prior pixel-level or global deep reinforcement learning (DRL) approaches, this work pioneers an object-centric decision-making framework. The task of selecting representative image patches is modeled as an environment for multiagent system interactions suitable for DRL. ObjectAgents consists of two agents: a PositionFilterAgent (PFA) and a ScaleControlAgent (SCA). These two agents work together to complete the classification process by observing the state of the environment and taking their own actions. The PFA is responsible for selecting optimal candidate points in the image for classification, while the SCA identifies the ideal image scale for accurate object identification. By using a learning mechanism where the agents receive rewards based on how much their actions improve the classification performance, the framework optimizes and learns the best decision-making strategies through DRL, enhancing the decision-making process and leading to better generalization in complex remote sensing scenarios. Experiments on IowaNet (1 m), the ISPRS Vaihingen benchmark (9 cm), and a constructed Global Tidal Wetlands dataset (10 m) demonstrate that ObjectAgents consistently outperforms manually designed (MD) baselines and remains robust in complex heterogeneous regions.**

***Index Terms*—Agent, object classification, position- and scale-aware, reinforcement learning, remote sensing.**

I. Introduction

REMOTE sensing classification remains a prominent and complex field within remote sensing image processing. The creation of land-cover maps has significantly enhanced our understanding of Earth's terrestrial environments, providing essential insights for natural resource management and land-use planning [1], [2]. With the advancement of Earth observation technology, the capability to acquire remote sensing data has been unprecedentedly enhanced [3], [4]. However, the intelligent processing of this vast amount of remote sensing data has become an urgent research problem that needs to be addressed [5], [6].

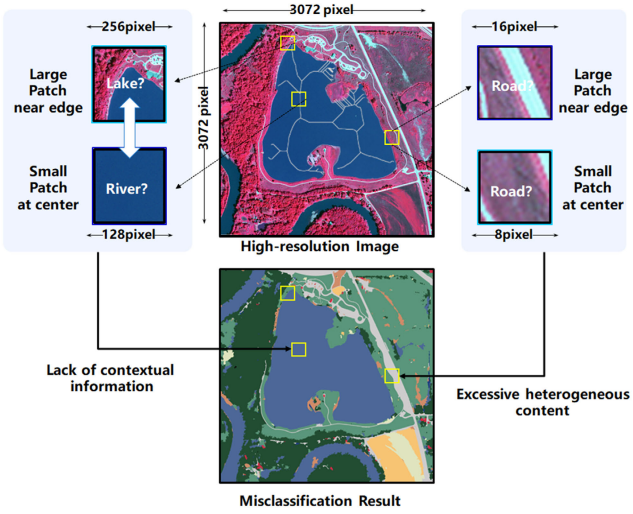


Fig. 1. Visual illustration of the impact of image block sampling on accurate object representation. The current methods often rely on manually set rules, directly selecting the geometric center or predefined scale, making it difficult to adaptively choose sampling regions based on object categories. The yellow boxes represent the sampled image blocks, and the white lines indicate the manually defined center lines of the sampling.

In recent decades, deep learning has emerged as a dominant approach for remote sensing image classification, owing to its robust end-to-end feature learning capabilities [7]. Fully convolutional networks (FCNs) [8] enhance performance by assigning categories to all pixels simultaneously [9]. However, as the model complexity increases, in models such as the vision transformer (ViT) and Swin transformer, the demand for extensive training samples grows exponentially [8], [10]. The diverse and complex distribution of objects in remote sensing images necessitates expert annotation, leading to high labor costs and often resulting in blurred boundaries [11].

Therefore, the deep object classification methods, which combine a deep learning classifier and image segmentation, have been proposed to enhance the classification accuracy [12]. By segmenting the image into homogeneous objects, the deep-

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learning methods enable the classifier to assign categories based on the features of the objects. This approach reduces redundancy and captures the spatial relationships, improving land-cover classification and robustness in complex remote sensing images [12], [13].

In the deep object classification framework, sampling representative image patches is a critical step for accurately representing objects [14], [15]. Correct image patch positions demonstrate a superior discriminative ability [16], while appropriate image patch scales allow the model to comprehensively use contextual information [17]. In research on position sampling, positions have been modeled using morphological and geometrical methods [18]. In research on scale determination, numerous studies have proposed various manually designed (MD) scale control methods [19]. Previous studies have shown that different scales and positions significantly impact the accuracy of classification [20], [21], [22]. Despite ongoing efforts, traditional handcrafted methods—such as selecting geometric centers, skeleton lines, or fixed grids—assume intra-object semantic consistency, which often breaks down in complex scenarios [19]. For example, irregular water bodies and curved road segments exhibit ambiguous boundaries, making center-based sampling unreliable. Likewise, fixed-scale windows fail to adapt to the contextual needs of different object types. These limitations are visually illustrated in Fig. 1, where using the geometric center or a predefined scale often leads to semantic omissions: patches centered on lakes miss edge information, while overly large patches in homogeneous regions like grassland introduce irrelevant background content—both contributing to misclassification. These challenges highlight the need for a class-aware and context-sensitive patch sampling strategy. This motivates the ObjectAgents framework, which learns to jointly optimize sampling position and scale in a data-driven manner [23], [24], [25].

To address the above adaptive selection problem, a position- and scale-aware object-agent classifier via deep reinforcement learning (DRL) for remote sensing imagery—named ObjectAgents—is proposed in this article. DRL, which is based on agent-environment interaction, enables the model to learn and make decisions through environmental rewards [26], [27]. This allows the model to autonomously and dynamically adapt actions without direct labels, thereby optimizing the performance in complex scenarios. DRL has been successful in various domains and appears promising for dynamically adjusting receptive field scales in land-use and land-cover segmentation networks [28]. However, prior work has primarily focused on semantic segmentation with pixel-level annotations, whereas our study addresses a fundamentally different problem—object-level classification. In this setting, each image patch, regardless of scale, is assigned a single label at its center as the class of the whole image patch, introducing new challenges for spatial sampling and contextual representation. Building upon previous insights into scale-adaptive decision making, this study develops a DRL-based framework tailored for object-level classification. This model uses the dynamic programming methods common in DRL to address the scale- and position-related issues in remote sensing image classification, achieving superior results compared to MD rules.

The contributions of this article are as follows.

1) *ObjectAgents:* A DRL-Based Position- and Scale-Aware Classifier. In ObjectAgents, the static classification process is represented as a scale-dynamic decision-making sequence, by the PositionFilterAgent (PFA) and ScaleControlAgent (SCA). The PFA selects the most discriminative points from a uniformly distributed set within the object to identify key locations. Meanwhile, the SCA determines the optimal image scale around these points to capture better contextual information for object classification.
2) *PFA:* The PFA uniformly distributes sample points within the object and uses a pretrained classification network to extract multiscale features. It predicts whether each sample point can be correctly classified across most scales, and receives rewards for successful predictions, enabling it to dynamically identify key representative points.
3) *SCA:* The SCA extracts multiscale and morphological features from the filtered sample points and fuses them via a cross-attention mechanism. Based on the fused contextual information, it selects the optimal image scale and learns through rewards derived from the classification accuracy feedback of a pretrained multiscale classifier.

Experimental evaluations on IowaNet [18], the ISPRS Vaihingen benchmark, and a Global Tidal Wetlands dataset [29] constructed from publicly available samples demonstrate the effectiveness of the proposed approach. The remainder of this article is organized as follows. Section II introduces the related works. Section III provides an introduction to the fundamental concepts of DRL. Section IV details the proposed ObjectAgents framework. Section V presents the experimental results and analysis. Finally, Section VI concludes the article.

II. Related Work

A. Remote Sensing Classification Based on Deep Learning

In recent years, deep learning has become the mainstream approach for remote sensing image classification, with numerous methods from different viewpoints. These methods can be roughly divided into three main categories: 1) patch-based classification methods; 2) dense pixel-based classification methods; and 3) deep object classification methods. By employing a sliding window process, the patch-based convolutional neural network (CNN) architecture leverages contextual information for class prediction. Initially introduced for urban mapping, this approach has subsequently been used in numerous other studies, including the generation of wetland inventory [30], [31] and the evaluation of a patch-based CNN for wetland mapping with a higher overall accuracy (OA) [32]. The dense pixel-based classification methods address this issue by predicting a class for each pixel of the original image and are now widely used in remote sensing. Nonetheless, these methods often require extensive manual annotation and can generate blurred edges during land-cover classification.

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Recently, deep neural network models have been widely investigated for addressing remote sensing classification tasks, reporting impressive performances [33]. A notable advantage of the deep-learning methods over the traditional approaches for land-cover description is their ability to adaptively learn distinguishing features from images in an end-to-end manner [34]. However, these methods often require extensive manual annotation of densely labeled pixels to obtain the ground truth, and are prone to generating blurred edges when performing land-cover classification [35], [36].

B. Object-Based Deep-Learning Methods for Remote Sensing Image Classification

In recent years, CNNs have been integrated into object-based image analysis (OBIAs) frameworks to achieve more precise object boundaries [13]. The object-based deep learning classification methods can be divided into three main steps. First, the image is segmented into several segments. Then, for a certain object, multiple positions are generated based on MD rules. Finally, the image patches centered at these positions are classified and labeled using deep learning to achieve object classification. Improvements on this basis can be divided into two main parts:

1) Image Patch Position Sampling: These methods investigate the role of different sampling strategies for object identification. For instance, Zhang et al. [13] determined the position of image patches by calculating the minimum bounding rectangle (MBR) of the object and the positions of its major and minor axes. Building on this, Huang et al. [37] and Martins et al. [18] proposed using skeleton line analysis to confirm image patch positions along the skeleton lines. Chen et al. [16] classified the entire object by assigning it the most frequent predicted class of random points within it to reduce misclassification. Lv et al. [17] introduced a morphology-based binary tree sampling (BTS) method to select optimal points, while Tong et al. [22] identified objects covered by multiple patches based on the patches that intersected with the largest area proportion.

2) Image Patch Scale Selection: These methods integrate multiscale information or use multiscale models to effectively mitigate misclassification at a single scale. Martins et al. [18] employed CNN classifiers at six different scales to select various image patch sizes within objects. Behera et al. [20] proposed a superpixel-aided multiscale network, using these information-rich superpixel images to extract scale-invariant features for prediction. Lv et al. [38] developed a region-based multiscale majority voting classification approach for high-resolution images. Zhang et al. [39] developed a method to transfer information through progressively larger scales, integrating small-scale details into broader contexts for land-cover and land-use learning. Wang et al. [21] proposed an object-scale adaptive CNN that combines OBIAs and deep learning, enhancing the classification accuracy through multi-scale analysis and majority voting. Tong et al. [22] improved the classification accuracy by segmenting the image into different-scale patches and using weighted fusion of samples to fully exploit the feature information at various scales.

While these methods enhance object classification outcomes by MD rules for selecting image scales and positions, they often struggle to simultaneously incorporate morphological information and land-cover categories, making it challenging to flexibly and adaptively select appropriate scales and positions based on land-cover types.

III. DRL

A. Agent System

An agent is a system that can autonomously perceive its environment, make decisions, and execute actions. Agents interact with their external environment to accomplish predefined tasks. This approach has been widely applied to model complex phenomena in various fields, such as autonomous driving [40], gaming [41], and automation [42].

Specifically, as an autonomous entity, an agent Ag can be represented by a tuple (S, A, R), where S, A, and R denote the agent's ***state***, ***action set***, and ***reward set***, respectively. Once the agent is activated, its percepts information from the ***environment*** E_i. This information assists the agent in combining its ***state*** s_i to select an appropriate ***action*** a_i and receive a reward r_i. In turn, the agent's action impacts the environment E_i.

B. DRL-Based Dynamic Networks

DRL is a method for training agents. In DRL, the agent continuously interacts with its environment to learn how to make decisions, with the goal of gradually improving its behavior through a process of trial and error. DRL not only helps agents to adjust themselves in uncertain and dynamic environments, but also guides them toward more optimal actions through a feedback mechanism involving rewards and penalties.

DRL is a highly flexible approach that commonly modeled as a Markov decision process (MDP) defined by the tuple (S, A, P, R) [43]. At each ***timestep*** t, the agent observes the current ***state*** s_t and selects an ***action*** a_t based on a ***policy*** $\pi(a|s)$. The environment then transitions to $s_{t+1} \sim P(\cdot|s_t, a_t)$ and yields an immediate reward $r_t = r(s_t, a_t, s_{t+1})$ [44]. During the agent training process, the sequence of states, actions, and rewards forms a complete ***trajectory*** τ. Training seeks a policy π that maximizes the expected discounted return ***cumulative reward*** $J(\tau)$.

In the context of CNNs, combining DRL can dynamically adjust the network architecture, either by modifying the network architecture based on the input or by optimizing the network performance through changing the input itself [45]. DRL has been widely applied in fields such as path planning [46], video games [47], and strategy games [26], often surpassing human performance and demonstrating powerful potential [48]. In the field of remote sensing, Zhong et al. [49] used a multiagent model to perform segmentation and classification of high-resolution remote sensing images. Cui et al. [50] employed spatio-temporal context, object appearance models, and motion vectors to provide occlusion information, driving DRL to achieve object tracking under complete occlusion. Liu et al. [51] proposed a knowledge-guided model for active

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object tracking, using the correlation between states and image quality as prior knowledge to guide the DRL algorithm in improving the observed image quality. In addition, Liu et al. [28] transformed the static patch size remote sensing image classification problem into a dynamic-scale decision-making process, effectively handling and providing feedback for various scales in land-cover classification. These studies have highlighted the revolutionary potential of DRL in dynamically improving network operations across various fields, making it a crucial tool for enhancing the efficiency and accuracy of automated systems.

IV. Methodology

To select the most representative image patch from an object for classification, ObjectAgents is designed to perform intelligent processing during the classification process, providing suitable image positions and scales. ObjectAgents comprises three sub-networks: the classification network $\pi_S(Y|X)$, the PFA $\pi_{pos}(a_{pos}|s_{pos})$, and the SCA $\pi_{scl}(a_{scl}|s_{scl})$. The overall workflow of the proposed Object Agents framework is shown in Fig. 2. In this framework, the aim is to identify the most representative image patches from segments of remote sensing images for classification. The process primarily involves filtering candidate points and selecting patch scales based on their geometric and categorical features, and it is modeled as an MDP [52].

In this work, the object-level patch selection process is defined as two episodic MDPs. For each segmented object ω with a set of candidate points $C(\omega) = \{p_1, \ldots, p_K\}$, we create one episode of length K. The environment consists of the input image, the object mask, the discrete scale set $S = \{S_1, \ldots, S_T\}$, and a frozen multiscale classifier ϕ (feature encoder E_{cls} and per-scale heads). States at different timesteps are i.i.d. conditioned on ω; the transition is, therefore, simple "advance-to-next-point." We use undiscounted returns with $\gamma = 1$ since episodes are short and finite.

The PFA $\pi_{pos}(a_{pos}|s_{pos})$ evaluates the confidence score P_{conf} of a point by analyzing the number of correctly identified scales among the candidate points using the multiscale features f_{pos} extracted from state s_{pos}. It receives rewards r_{pos} for correctly selecting points and excluding problematic ones. Concurrently, the SCA $\pi_{scl}(a_{scl}|s_{scl})$ extracts multiscale features from contextual information $\{f_i|i = 1, 2, \ldots, S_{\max}\}$ and geometric features g, which are integrated using a cross-attention mechanism M_{att}. These features serve as the crucial information learned by the agent from the observations. The SCA dynamically selects appropriate image scales based on the various geographical features and land-cover categories of the candidate points, and gets rewards r_{scl} if it selects the correct one.

The purpose of ObjectAgents is to train both agents to provide suitable candidate points and a suitable image scale for classification, thereby maximizing the expected cumulative reward $J(\tau)$.

To ensure stable learning, ObjectAgents adopts an alternating strategy that combines supervised learning and DRL. The classification network is first pretrained on multiscale patches with ground-truth labels and then fixed to serve as the environment for the PFA and SCA. These agents are trained sequentially using proximal policy optimization (PPO): PFA filters high-confidence points, and SCA selects optimal patch scales. This cooperative design enables class-aware and context-sensitive sampling decisions.

A. Development of Object-Based Classification Models

1) Object-Based Image Segmentation: Image segmentation serves as a fundamental preprocessing step in the proposed approach. The mean shift algorithm [53] is used, which is renowned for its robust edge detection capabilities in high-resolution imagery. This OBIA algorithm produces segmented images that maintain homogeneous spatial and spectral characteristics derived from the input data. Each image X in the dataset is segmented into a set of objects $\{(o_1, o_2, \ldots, o_{N_{obj}})\}$, which are used as inputs for the subsequent analysis. The segmentation parameters—spatial scale (Hr), spectral scale (Hs), and minimum segment size (Ms)—are meticulously calibrated based on both quantitative and visual evaluations, aligning with the methodologies adopted from the multiscale object-based CNN (multi-OCNN) approach [18]. These parameters ensure the preservation of the critical image details necessary for the subsequent analytical processes.

2) Establishment of Multiscale Classification Models: In this article, the multiscale concept is defined as using input images with different patch sizes in the classification model. Employing a multiscale classification model, which predicts the classes of real-world objects using various image patch sizes, can enhance the model's performance in understanding the contextual information of the object types [54]. This capability is achieved by training multiscale ResNet [55] classifiers $\pi_S(Y|X)$ across a range of scales $S \in \{S_1, S_2, \ldots, S_{\max}\}$ for the object-based classification. In the training process, sample points with labels Y_i are used as centers to generate a series of image patches $\{x_i \in \mathbb{R}^{H\times W\times C}|i = 1, 2, \ldots, S_{\max}\}$ at different scales. These patches serve as the inputs for training the classifiers $\pi_S(Y|X)$ at their corresponding scales S. in addition, a multilayer perceptron (MLP) classifier [56] is employed to categorize individual pixels, aiding in delineating object edges and further enhancing the classification accuracy in complex scenes.

3) Object-Based Classification Workflow: The trained classification model computes the per-class probabilities for the representative image patches selected from the object X_i^S and assigns the object class $\widehat{Y_i}$ based on the highest score. For an image with N_{obj} objects, each having N_{pts} convolutional positions, the model predicts an n-dimensional probability vector at each position. These vectors indicate the membership probability for each class $n \in \{1, 2, \ldots, N_{cls}\}$. The final class label for each object is determined by the highest cumulative score across all the vectors

$$\vec{P}_i = [c_1, c_2, \ldots, c_n] = \sum_{n=1}^{N_{cls}} c_n = 1; \quad 0 \le c_n \le 1 \tag{1}$$

$$C = \operatorname{argmax}\left(\sum_{i=1}^{N_{pts}} \vec{P}_i\right). \tag{2}$$

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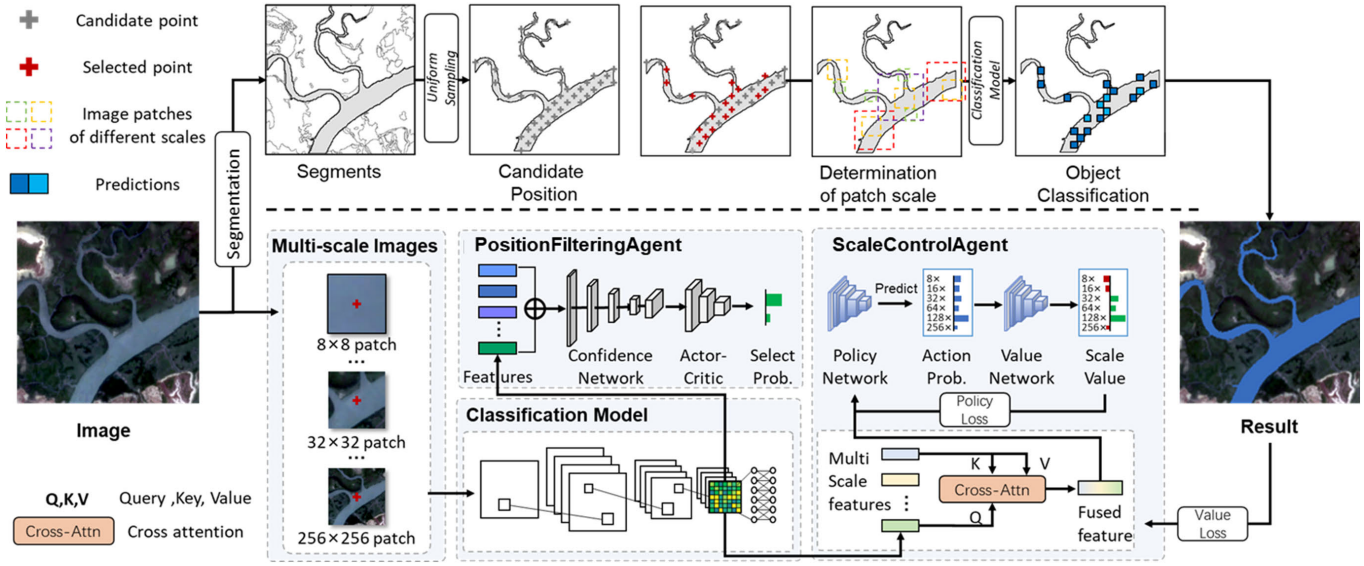


Fig. 2. Overall flowchart of the proposed DRL-based scale- and position-adaptive ObjectAgents network.

This classification procedure is applied to all the image objects. Within the process, the selection of image patches is a core component of this method, which is accomplished through the proposed ObjectAgents. ObjectAgents consists of two main components: the PFA and the SCA. Initially, candidate points are uniformly sampled from within the object and refined by the PFA, which selects high-confidence patches. The SCA then adjusts the image patch sizes for optimal classification.

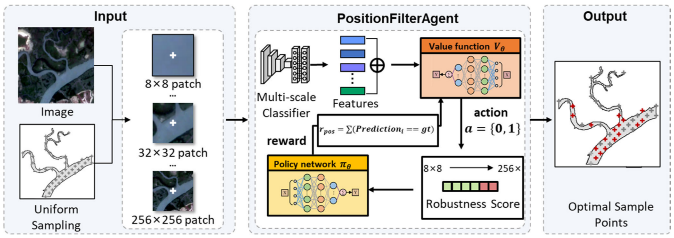


Fig. 3. Flowchart of the PositionFilterAgent.

B. PositionFilterAgent

In our approach, the PFA $\pi_{pos}(a_{pos}|s_{pos})$ improves classification by making a binary keep/discard decision for each candidate point, especially near object boundaries that are prone to confusion. We cast this as an object-level episodic MDP $\mathrm{M}_{pos} = \{S_{pos}, A_{pos}, R_{pos}, \tau_{pos}, \gamma\}$: for each segmented object ω with candidates $C(\omega) = \{p_1, \ldots, p_K\}$, one episode of length K deterministically advances from p_t to p_{t+1} and terminates at $t = K$ (finite horizon; $\gamma = 1$). At each timestep t, the state concatenates multiscale features of patches centered at the current point—extracted by a frozen multiscale classifier ϕ—with a positional encoding of normalized distance-to-boundary; the action $a_{pos} = \{0, 1\}$ sampled from A_{pos} is a binary keep/discard choice; the transition moves to the next candidate in a fixed order. Reward, in brief, encourages retaining points whose cross-scale predictions are consistent and discarding ambiguous or scale-inconsistent points; the opposite decisions are penalized. The exact mathematical form, thresholds, and hyperparameters of the reward are provided in Section IV-B [(5)].

The goal of the PFA is to maximize the cumulative reward $J(\tau)$ for the sequence within each object. In addition, in the proposed approach, the classification process is established as discrete, with the state-action pairs of each point being independent of each other. The initialization of candidate points is achieved through uniform, equidistant sampling within the object.

The PFA uses the encoder $\phi_S(X)$ from a pretrained classification network $\pi_S(Y|X)$ to extract features $\{f_i | i = 1, 2, \ldots, S_{\max}\}$ from multiscale image patches centered on the candidate points, which represent the state s^t_{pos}. The extracted features are then flattened and concatenated to serve as the input for the agent's decision-making process. The training of the PFA is grounded on the premise that points which can be correctly classified across more scales are considered to be of higher confidence.

The agent includes a policy network θ^{pos}_A and a value network θ^{pos}_C as its core components as shown in Fig. 3. The policy network θ^{pos}_A determines the appropriate action for each sample point P_t within the object. The action space is defined as $a^t_{pos} \in \{0, 1\}$, where the actions represent the decision to either discard the point or retain it as representative of the object, thereby enhancing the classification network's confidence and accuracy. The value typically refers to the expected cumulative reward that an agent can obtain from a certain state or state-action pair (s_{pos}, a_{pos}). The value network in this context refers to the estimation of the expected cumulative rewards for given state-action pairs. By evaluating the value of different states or state-action pairs, the value network θ^{pos}_C provides guidance for improving the policy network by indicating the magnitude of the rewards r_{pos} received. This process reduces the variance in gradient estimation, thereby making the training process more stable and efficient.

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To enhance the robustness of the PositionFilterAgent (PFA), we adopt an automated, fixed sampling prior that systematically biases candidate selection toward object boundaries—regions that are inherently susceptible to misclassification.

Specifically, we employ a soft 2-D Gaussian mask centered at the patch midpoint to assign higher sampling probabilities to boundary regions. Thresholding this mask then defines the candidate regions to further generate boundary-biased candidate points. The 2-D probability map is formulated as

$$Z(x,y) = \exp\left(-\frac{(x-x_c)^2+(y-y_c)^2}{2\sigma^2}\right) \tag{3}$$

where (x_c, y_c) denotes the center of the patch and σ represents the standard deviation in normalized coordinates. A binary mask is then constructed as

$$Mask(x,y) = \begin{cases} 1, & \text{if } Z(x,y) \geq \tau \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where τ controls the spatial focus sharpness. In cases where the resulting region is insufficiently large, uniform sampling within the object area serves as a fallback mechanism. This strategy is entirely automatic, consistent across samples, and effectively exposes the PFA to more ambiguous regions—facilitating better generalization.

Regarding the optimization objective, the agent is rewarded for selecting points that demonstrate classification consistency across the majority of scales. Conversely, it incurs a penalty for selecting low-confidence points or discarding valid ones. The reward function $R_{pos}(at, y)$ is expressed as follows:

$$R_{pos}(a_t, y) = 2 \times \left| \frac{\sum_{S=1}^{N_{scale}} 1(\widehat{y}_S == y_S)}{S_{\max}} - 1 \right| \times (at - 0.5) \tag{5}$$

where $\widehat{y}_S$ predicted class label at scale S, and y_s is the ground-truth class label, which remains invariant across scales. The indicator function $1(\widehat{y}_S == y_S)$ returns 1 if the prediction matches the ground truth at scale S, and 0 otherwise. N_{scale} represents the total number of evaluated scales. The agent's action at timestep t is denoted by $a_t \in \{0, 1\}$, where $a_t = 1$ indicates that the candidate point is retained, and $a_t = 0$ indicates that it is discarded. This reward structure incentivizes the retention of points with high multiscale consistency while penalizing the retention of ambiguous points, ensuring that the agent's decisions align with robust classification performance.

C. ScaleControlAgent

The sampled points p' within each object o that pass the PFA $\pi_{pos}(a_{pos}|s_{pos})$ are treated as high-confidence inputs for scale selection, which we also cast as an episodic MDP with one episode per object and one timestep per retained point. Formally, the SCA is defined as $M_{scl} = \{S_{scl}, A_{scl}, R_{scl}, \tau_{scl}, \gamma\}$. For a PFA-retained point p',the state $s_{sca}(p') \in S_{sca}$ fuses multiscale features extracted by the frozen backbone encoder E_{cls} with a positional encoding of the normalized distance to the object boundary via a lightweight cross-attention module (details in Section IV-C).

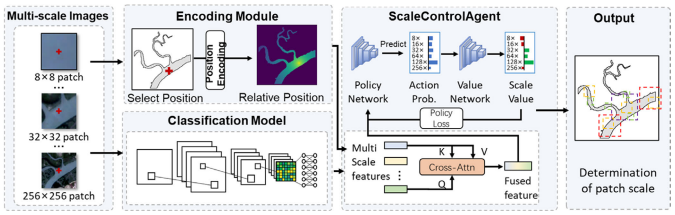


Fig. 4. Flowchart of the ScaleControlAgent.

The action space A_{scl} is the discrete scale set $A_{scl} = \{S_1, \ldots, S_T\}$; at each timestep the agent selects a single scale. Reward, in brief, encourages choosing scales that yield correct classification while remaining compatible with local geometry (e.g., not excessively larger than the point-to-boundary distance); the exact mathematical form and hyperparameters are given in Section IV-C [(10)]. The transition deterministically advances to the next retained point p' and terminates after all are processed; we use a finite horizon with $\lambda = 1$. The SCA is trained to maximize the cumulative return over the per-object sequence.

Specifically, to simulate the multiscale features $\{f_i | i = 1, 2, \ldots, S_{\max}\}$ around point p' when observing objects, the multiscale image patches centered around candidate points are encoded as part of the overall feature set using an encoder $\phi_S(X)$ that is separated from the pretrained classifiers $\pi_S(Y|X)$ as shown in Fig. 4. In addition, positional encoding X_{PE} is performed based on the distance from the point to the object's edge, with values outside the object's edge set to 0. To extract the relative positional information feature g, an encoder G is employed to extract the feature from X_{PE}. Subsequently, the relative position feature g and the multiscale features $\{f_i | i = 1, 2, \ldots, S_{\max}\}$ are fused using a simple multiheaded cross-attention mechanism M_{att}.

Given three common sets of inputs—a query set Q, a key set K, and a value set V—we define the multiheaded cross-attention mechanism M as follows:

$$M(Q,K,V) = \left(\Gamma_1 \oplus \Gamma_2 \cdots \oplus \Gamma_{N_\Gamma}\right) W \tag{6}$$

$$\Gamma_\mu = A\left(QW_\mu^Q, KW_\mu^K, VW_\mu^V\right)\Big|_{\mu=1}^{N_\Gamma} \tag{7}$$

$$A(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_f}}\right) V \tag{8}$$

where Γ_μ is the μ-th head, and A denotes the attention function. W, W^Q, W^K, W^V are learnable linear mappings, and d_f is a scaling factor. The multiscale fusion can then be formulated as

$$\varphi = M\left(g, \left\{f_i|_{i=1}^{N_{scale}}\right\}, \left\{f_i|_{i=1}^{N_{scale}}\right\}\right) \tag{9}$$

where φ is the fused feature used as the input for the agent's decision-making process. $\{f_i|_{i=1}^{S_{\max}}\}$ is feature vectors extracted from the candidate point P_t at scale i typically from an encoder or classifier. g is the positional embedding that encodes the normalized distance from the candidate point to the object boundary, offering spatial context. is a multihead cross-attention module that fuses the features. $M(\cdot)$ Its internal mechanism is defined in (6)–(8).

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This formulation treats the position feature g as the query, and the multiscale features $\{f_i|_{i=1}^{S_{\max}}\}$ as keys and values in a multihead attention mechanism. The attention computes semantic relevance between the candidate point's location and the feature representation from each scale. Through learned attention weights, the module fuses relevant features, emphasizing the most semantically meaningful scales.

The core components of the SCA $\pi_{scl}(a|s)$ are similar to those of the PFA, comprising a policy network θ_A^{scl} and a value network θ_C^{scl}. The policy network takes appropriate actions a_{scl}^t at each timestep t based on the state s_{scl}^t. The action space A_{scl}, denoted as $a_{scl}^t \in \{1, 2, \ldots, S_{\max}\}$, represents the different-scale image patches selected by the SCA centered at the image point, where the smallest scale is a single pixel. The value network evaluates the action-state function of the agent based on the designed reward function R_{scl} and guides the update of the policy network.

During training, the global image label is assigned to the center object. The selected-scale image patches are subsequently processed by corresponding scale classifiers and compared with the label Y. Specifically, the smallest scale (1×1) uses an MLP, while larger scales employ deep classifiers. Fundamentally, probabilistic predictions $\widehat{Y}$ are aggregated to formulate the reward, where misclassifications are penalized. However, given the high-dimensional state space, this naive reward often fails to converge in complex scenarios. Therefore, the proposed approach employs a two-phase reward setting to guide the agent in learning the correct scale selection strategy. In the first phase, MD rules teach the agent basic scale selection, rewarding it only when its choices match the rules. In the second phase, the agent explores freely, receiving rewards for correctly classified scales. The specific reward function is as follows:

$$R_{scl}(a_t, y) = \begin{cases} 1, & \text{if } t \le k \text{ and } a_t = S_m \\ -1, & \text{if } t \le k \text{ and } a_t \ne S_m \\ \dfrac{S_{\max} - (a_t - \min(I))}{S_{\max}}, & \text{if } t > k \text{ and } \widehat{y} = y \\ -\dfrac{\sum_{S=1}^{S_{\max}} (\widehat{y}_S = y_S)}{S_{\max}}, & \text{if } t > k \text{ and } \widehat{y} \ne y \end{cases} \tag{10}$$

where x_S, y_S represent the image patch of scale S and its label, respectively. $\widehat{y}_S = \pi_S(x_S)$ is the prediction of x_S by classifier $\pi_S(Y|X)$. $I = \min(\text{argwhere}(\widehat{y}_S = y_S))$ is the minimum scale, where y_S equals $\widehat{y}_S$. In (10), k is a user-defined timestep experience parameter.

Since training scale variation is limited, the SCA may be biased toward consistently larger scales. To counteract this, penalties are imposed if the selected scale exceeds the sample-to-boundary distance. Additionally, data augmentation via random resizing (factors 0.5–1.25) and padding is applied. This strategy effectively prevents the agent from overfitting the selection of disproportionately large scales.

D. Optimization of ObjectAgents

To achieve stable learning of the agents in an adaptive environment, the PPO algorithm is employed [57]. As previously described, the SCA and PFA modules consist of two learnable components: an "actor" network θ_A that learns the parameterized policy, and a "critic" network θ_C that learns the value function to evaluate the state-action pairs. The critic provides reinforcement signals to the actor by learning the advantage function $A_\pi(s, a)$ [58]

$$A_\pi(s, a) = Q_\pi(s, a) - V_\pi(s) \tag{11}$$

where $V_\pi(s)$ denotes the cumulative reward from using policy π starting from state s, and $Q_\pi(s, a)$ denotes the cumulative reward $J(\tau)$ obtained by performing action a from state s under policy π.

The target advantage value $V_{tar}^\pi(s_t)$ can be estimated using the temporal difference algorithm and trajectory data, and the critic network θ_C is trained with the mean squared error (mse) loss

$$L_{value}(\theta_C) = \frac{1}{T} \sum_{t=0}^{T} \left(V_{tar}^\pi(s_t) - V_\pi(s_t)\right)^2. \tag{12}$$

The actor network predicts the advantage value $A_\pi(s_t, a_t)$ and optimizes it using the PPO-clipped surrogate objective function [57]

$$L_{policy}(\theta_A) = \frac{1}{T} \sum_{t=0}^{T} \min(r_t(\theta)\widehat{A}_i, clip(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\widehat{A}_t) \tag{13}$$

where $r_t(\theta)$ denotes the probability ratio of the current policy to the old policy, $\widehat{A}_t$ is the advantage estimate at time t, and ε is a hyperparameter, typically between 0.1 and 0.2, which limits the extent of the policy updates.

After candidate-point extraction, we first pretrain the classifier $\pi(Y|X)$ with fixed scales and freeze it as the shared environment. We then train the PFA with PPO; its reward is based on cross-scale prediction consistency, and it strictly discards ambiguous points (e.g., scale-inconsistent or boundary-confounded samples). Next, on the retained points, we train the SCA, whose reward combines classification correctness at the selected scale with a scale–to–boundary distance regularizer. The procedure follows a fixed PFA $\rightarrow$ SCA order and leverages the frozen classifier to enable indirect coordination between agents. This schedule is not end-to-end joint optimization, NOR within-epoch alternation; rather, it is a sequential curriculum that allows each agent to specialize in its subtask while their staged decisions jointly optimize position and scale selection, yielding stable and superior classification performance.

To provide a clear procedural view of the alternating training strategy and agent interaction, the complete training loop is detailed in Algorithm 1. This algorithm explicitly illustrates how the frozen environment interacts with the sequential optimization of PFA and SCA. We first pretrain and freeze the multiscale classifier ϕ (environment). We then optimize the PFA policy π_{Pos} and the SCA policy π_{Pos} with PPO in sequential epochs: π_{Pos} is trained on boundary-biased candidate points (Section IV-B), and π_{Scale} is trained on the retained points with fused multiscale features (Section IV-C). Rewards R_P and R_S are defined in Sections IV-B and IV-C, respectively.

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Algorithm 1 Training Loop of ObjectAgents

Input: Segmented object set ω; pretrained multi-scale classifier ϕ; discrete scales $S = \{S_1, \ldots, S_{max}\}$; sampling params $(K, \tau 1, \tau 2, Kmin)$; Epochs (E_p, E_S)

Output: Trained PFA policy π_{Pos}, trained SCA policy π_{Scale}

Freeze classifier ϕ
For epoch = 1 to E_p do: // Train PFA
  $D_p \leftarrow \emptyset$
  For each object $\omega \in \Omega$ do
    C1 $\leftarrow$ Sample K$(\omega, K; band\tau 1 \le d(x) \le \tau 2\})$
    For each point $p \in$ C1 do
      $s_{Pos} \leftarrow$ CrossScaleFeat + PosEnc(ϕ, p)
      $a_{Pos} \sim \pi_{Pos}(\cdot|s_{Pos}) \in$ {Keep, Ignore}
      $r_{Pos} \leftarrow$ R$_{pos}(a_{Pos}, p)$ // Equation (5)
      Store $(s_{Pos}, a_{Pos}, r_{Pos})$ into D_p
      End for
    End for
  Update π_{Pos} with PPO using D_p
End for
For epoch = 1 to E_S do: // Train SCA
  $D_p \leftarrow \emptyset$
  For each object ω in Ω do
    C2 $\leftarrow$ Points retained by π_{Pos}
    For each point p in C
      $s_{scl} \leftarrow$ FuseMultiScaleWIthPos(ϕ, p)
      $a_{scl} \leftarrow \pi_{Scl}(\cdot|s_{scl}) \in \{S_1, \ldots, S_{\max}\}$
      $r_{scl} \leftarrow$ R$_{\text{Scl}}(a_{scl}, p)$ // Equation (10)
      Store $(s_{scl}, a_{scl}, r_{scl})$ into D_S
      End for
    End for
  Update π_{scl} with PPO using D_S
End for

V. Experiments

A. Experimental Setup

All the experiments were conducted in PyTorch [59] and Stable Baselines3 [60]. The loss function employed was the cross-entropy loss function. The training optimizer for all the models was the Adam optimizer. The models were trained for 100 epochs, with an initial learning rate of 2e-4, which was subsequently decreased. The batch size was proportionally reduced based on the image block scale, with a minimum batch size of 96 for the largest image blocks and a maximum batch size of 3072 for the smallest image blocks. This resulted in six remote sensing image classification models of different scales. The experiments were run on two NVIDIA RTX A6000 GPUs (each with 48 GB GPU memory).

Segmentation and multiscale sampling. We first segment each scene with mean shift, using visually tuned parameters to preserve intra-object homogeneity (spatial radius [10], [15], spectral radius [10], [12]). Based on these segments, we sample multiscale patches from the set $\{8, 16, 32, 64, 128, 256\}$.

Reinforcement learning setup (PPO for PFA/SCA). We use PPO with standard settings unless noted: clip $\varepsilon = 0.2$, discount $\gamma = 0.99$, GAE $\lambda = 0.95$, rollout length $n_{steps} = 2048$, epochs per update $n_{epochs} = 10$, minibatch 64, LR 3×10^{-4}, entropy coef 0.0, value coef 0.5, max-grad-norm 0.5, and 200 k total steps per agent.

Datasets and evaluation. The classification assessment was conducted on three datasets (IowaNet, ISPRS Vaihingen, and Global Tidal Wetlands). For the IowaNet dataset, a two-stage stratified random sampling was used [61]. Initially, 1250 points were randomly sampled in each region. To ensure adequate representation of the rare classes, the second stage guaranteed at least 125 samples per class, resulting in 5000 samples as reference data (625 samples per class). For the Global Tidal Wetlands dataset, random sampling was used to select 6000 points from the 40 885 available points as reference data (1000 samples per class). For the ISPRS Vaihingen dataset, we constructed a multiscale, center-aligned object-classification subset by sampling multiple points per connected-component object and extracting point-centered multiscale patches; each patch inherits the sampled-point label. We also crop a central 8×8 label window per patch to supervise auxiliary segmentation.

In this study, the classification performance was evaluated using the OA, Kappa coefficient (κ), and user's accuracy (UA), along with a confusion matrix to analyze agreement and disagreement. OA measures the proportion of correctly classified instances, κ accounts for chance agreement, and UA reflects the accuracy of pixels assigned to a specific class. Together, these metrics provide a comprehensive assessment of classification performance.

B. Experiment 1: IowaNet Dataset

IowaNet is a multiscale land-cover classification dataset created from high spatial resolution (1-m) National Agriculture Imagery Program (NAIP) 2015 aerial imagery. It comprises 1 000 000 images across 10 land-cover classes, with 100 000 images per class. These classes include structure, road, river, lake, cropland, fallow, shadow, forest, grassland, and barren. Examples of the land-cover classes in the IowaNet dataset are shown in Fig. 5. To enhance the CNN performance, rivers and lakes were separated due to their distinct geometries and sizes.

For this experiment, we used the false-color bands of IowaNet (green, red, near-infrared), providing a rigorous benchmark for land-cover classification. The dataset was divided into training, validation, and test sets at a ratio of 9:1. From the training set, 200 000 images were randomly selected for the DRL training.

To substantiate the superiority of the proposed ObjectAgents method, we conducted a comprehensive comparison against a diverse set of baselines, ranging from classical strategies to recent state-of-the-art (SOTA) deep learning architectures. These comparisons are categorized as follows.

SOTA Deep-Learning Models: We included SegFormer [63], a representative transformer-based framework, to evaluate performance against self-attention mechanisms. Additionally, we compared our method with ConvNeXt-UPerNet [62], which represents the current SOTA in modern CNN architectures.

Established Multiscale Methods: We compared against classical multiscale classification approaches, including multi-OCNN [18], fixed-scale OCNN [13], and OSA-CNN [21].

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TABLE I
COMPARISON OF DIFFERENT LAND-COVER CLASSIFICATION METHODS ON THE IOWANET DATASET

Method	IowaNet											
	Kappa	OA(%)	Barren	Crop	Fallow	Forest	Grass	Lake	River	Road	Shadow	Structure
Fixed-scale patch-based methods												
Patch-based	0.7842	80.6	84.4	93.6	86.7	94.9	87.9	65.7	85.3	87.2	51.1	65.7
Fixed-scale	0.8102	82.9	88.9	95.0	90.5	84.8	95.8	84.8	86.9	86.3	59.2	56.1
ConvNeXt-UPerNet	0.8278	85.39	98.8	89.1	69.6	87.3	91.3	93.6	93.3	80.36	27.4	68.2
SegFormer	0.8456	87.0	88.7	89.1	80.6	87.3	90.1	96.7	93.2	76.5	23.7	74.6
BTS	0.8374	85.4	92.2	96.2	83.7	95.9	91.7	68.5	90.2	89.8	69.2	73.6
Multi-scale patch-based methods												
MOCNN	0.8676	88.9	91.7	95.5	94.3	92.9	96.7	88.6	84.4	91.5	69.8	74.3
PT-GID	0.8410	85.7	83.2	93.9	89.5	73.4	85.9	85.2	87.9	77.9	93.1	95.2
RMV	0.8346	85.9	90.6	94.5	87.3	87.2	94.1	83.2	84.0	91.1	59.4	77.9
OSA-CNN	0.8576	87.2	86.5	97.3	94.3	88.9	95.6	90.5	90.8	88.3	52.3	81.9
Position- and scale-aware method												
Proposed	0.8965	90.7	91.7	98.2	95.2	88.9	97.8	91.4	84.4	90.4	82.6	85.7

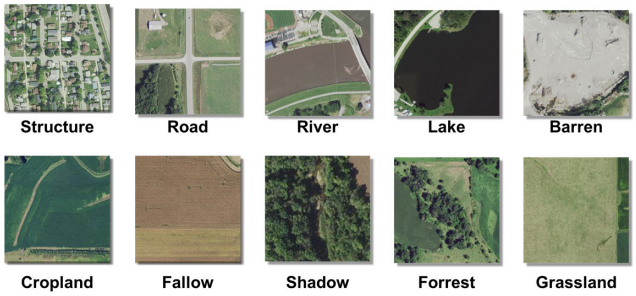


Fig. 5. Examples of the land-cover classes in the IowaNet dataset.

Sampling and Aggregation Strategies: We also evaluated region-based majority voting (RMV) [38] and position sampling methods such as BTS [17] and PT-GID [22] (adapted in simplified forms) to benchmark our agent-based sampling policy.

This diverse selection ensures that ObjectAgents is benchmarked against both established methodologies and cutting-edge Transformer/CNN architectures.

1) Qualitative Comparison: We compare ObjectAgents with a diverse set of baselines, including BTS (fixed-scale with varied positions), multi-O-CNN and OSA-CNN (multiscale object-based models), and SOTA segmentation models such as SegFormer and ConvNeXt-UPerNet, covering both traditional and modern approaches. Fig. 6 shows the classification results of the different methods on the IowaNet dataset. The IowaNet dataset contains narrow, context-sensitive features like rivers and pastures, where classification is highly scale-dependent. Fixed-scale and rule-based multiscale methods often misclassify these features due to their inability to adapt patch scale and position based on category. For instance, small patches centered on rivers may confuse them with lakes, while large patches at pasture edges may mistake them for roads. In contrast, ObjectAgents adaptively select scale and position, effectively capturing context and improving classification for such narrow, challenging features, as confirmed by visual results.

2) Quantitative Comparison: Table I presents the quantitative results of the different methods. The proposed method achieves an OA of 90.7%, significantly outperforming both the fixed-scale and multiscale methods. Among the fixed-scale methods, the BTS method obtains the highest OA, at 85.4%. In the multiscale methods, OSA-CNN achieves an OA of 87.2%, surpassing the other methods with manually selected scales. Furthermore, the proposed method demonstrates a superior classification accuracy across the various categories. For instance, in the crop category, ObjectAgents achieves a classification accuracy of 98.2%, compared to 97.3% for the OSA-CNN method, representing an improvement of 0.9%. In the lake category, the proposed method achieves a classification accuracy of 91.4%, compared to 90.5% for the OSA-CNN method, showing an improvement of 0.9%.

3) Position and Scale Sensitivity Analysis: As illustrated in Fig. 7, the confusion matrix comparison highlights the effectiveness of adaptive position and scale selection. Fixed-scale and fixed-position baselines exhibit significant confusion in categories like river, pasture, and shadow due to mismatched patch size or location.

In contrast, ObjectAgents produces clearer diagonal patterns, indicating improved discriminability—particularly for elongated or edge-aligned features. These categories are also common in fragmented or transitional landscapes, such as urban–rural edges and mixed agricultural landscape, suggesting strong transferability and scalability of our method in diverse remote sensing scenarios.

In spite of this, the performance on certain ambiguous or visually overlapping categories—such as shadow and structures—remains suboptimal. We attribute this to two potential factors: 1) limited spectral separability between shaded areas and nearby artificial surfaces and 2) insufficient semantic cues in local patches due to occlusions or mixed

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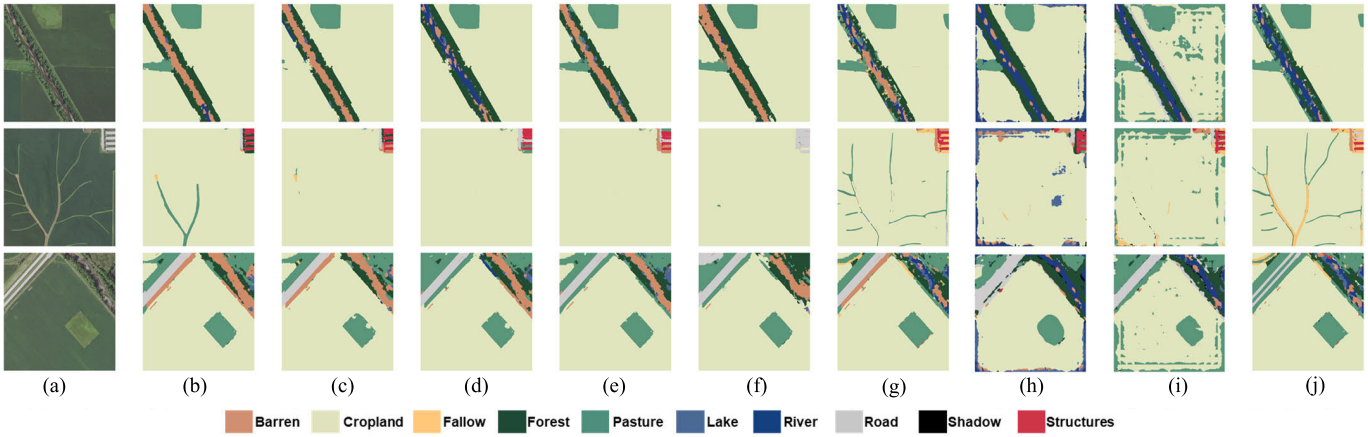


Fig. 6. Visual comparison of the classification results on the IowaNet dataset. (a) Image. (b) OCNN64. (c) BTS result. (d) RMV result. (e) PT-GID. (f) OSA result. (g) Multi OCNN. (h) ConvNeXt. (i) Segformer. (j) ObjectAgents.

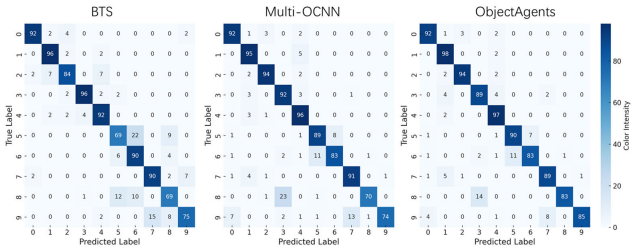


Fig. 7. Confusion matrices comparing classification performance under different strategies (left to right): Fixed-scale with varied positions, multiscale with fixed position, and ObjectAgents with adaptive position and scale selection.

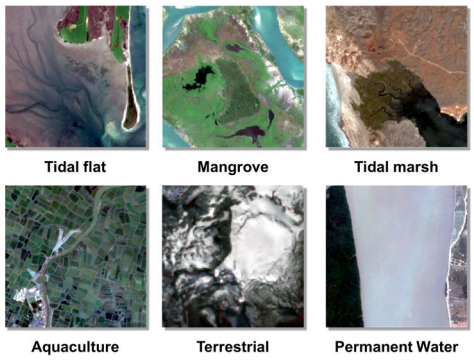


Fig. 8. Examples of land-cover classification with the Global Tidal Wetlands dataset.

boundaries. These limitations suggest that future extensions could incorporate broader temporal context or global priors to further enhance the classification of visually similar land types.

C. Experiment 2: Global Tidal Wetlands Dataset

In order to further evaluate the ability of the proposed ObjectAgents method to deal with object classification, we constructed a multiscale dataset based on publicly available global tidal wetlands sample points, comprising 40 885 accurately labeled points with corresponding Sentinel-2 10-m resolution multispectral imagery.

The dataset includes 23 138 wetland instances (tidal flats, tidal marshes, mangroves) and 17 747 nonwetland instances (aquaculture, terrestrial, permanent water), capturing various coastal features. Representative examples from the Global Tidal Wetlands dataset are shown in Fig. 8. The Sentinel-2 imagery includes visible (blue, green, red), near-infrared, and shortwave infrared bands, providing comprehensive spectral information for land-cover classification.

1) Qualitative Comparison: Fig. 9 illustrates the classification results of the various methods on the Global Tidal Wetlands dataset. The fixed-scale methods often misclassify the narrow tidal channels and confuse similar land-cover types, due to their inability to adapt the patch scales and positions. The MD multiscale methods show an improved performance but remain limited by the predefined rules, leading to misclassification in complex areas. In contrast, the ObjectAgents method adaptively selects optimal patch scales and positions, effectively capturing critical spatial and contextual information. This approach results in more accurate delineation of narrow features, better handling of transitions between land-cover types, and overall improved classification performance, particularly in regions with high feature complexity.

2) Quantitative Comparison: Table II presents the quantitative results of the different methods. The proposed method achieves an OA of 92.1%, significantly outperforming both the fixed-scale and multiscale patch-based methods. Among the fixed-scale methods, the BTS method obtains the highest OA, at 87.6%. Among the multiscale patch-based methods, the OSA-CNN method achieves an OA of 88.8%, surpassing the other methods with manually selected scales. Furthermore, the proposed method demonstrates a superior classification accuracy across the various categories. For example, in the tidal flat category, the proposed method achieves a classification accuracy of 83.1%, while the OSA-CNN method achieves 84.2%. In the mangrove category, both the ObjectAgents and OSA-CNN methods achieve an accuracy of 96.0%.

On the Global Tidal Wetlands dataset, ObjectAgents achieves the best OA and Kappa, outperforming all baselines. Its adaptive scale and position selection effectively reduce confusion in complex classes like tidal flats and aquaculture, where narrow or heterogeneous features are common. This highlights the importance of scale and position awareness for precise coastal land-cover classification.

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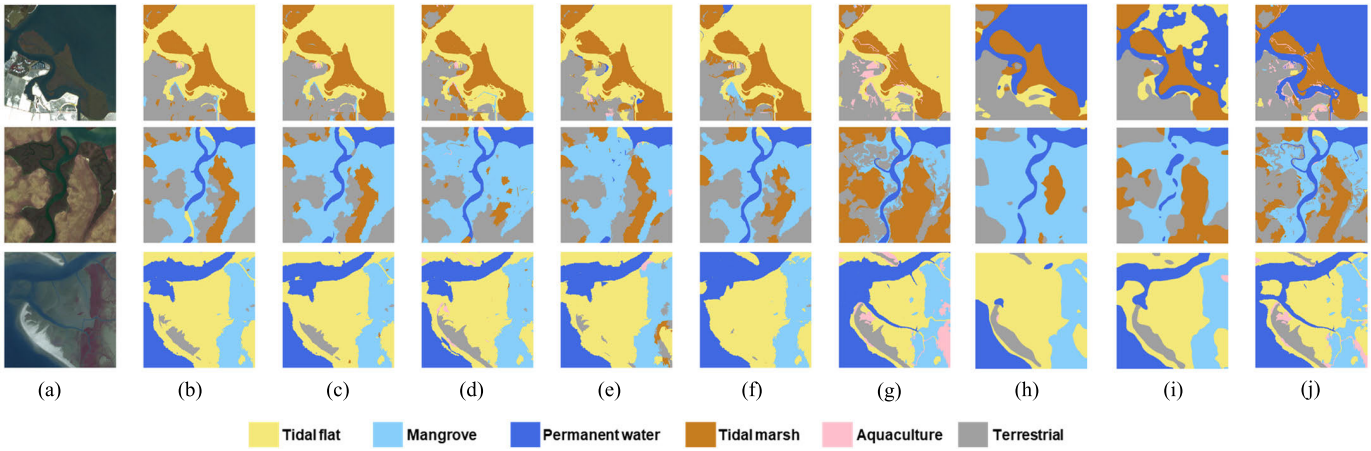


Fig. 9. Visual comparison of the classification results on the Global Tidal Wetlands dataset. (a) Image. (b) OCNN64. (c) BTS result. (d) RMV result. (e) PT-GID. (f) OSA result. (g) Multi OCNN. (h) ConvNeXt. (i) Segformer. (j) ObjectAgents.

TABLE II
COMPARISON OF DIFFERENT LAND-COVER CLASSIFICATION METHODS ON THE GLOBAL TIDAL WETLANDS DATASET

Method	Global Tidal Wetlands							
	Kappa	OA(%)	Terrestrial	Water	Tidal flat	Mangrove	Tidal marsh	Aquaculture
Fixed-scale patch-based methods								
Patch-based	0.7713	81.7	75.2	88.2	64.4	87.0	82.3	82.0
Fixed-scale	0.8502	88.1	77.9	86.3	86.4	95.7	90.2	91.2
ConvNeXt-UPerNet	0.8515	88.1	82.4	73.5	85.5	95.7	92.7	93.3
SegFormer	0.8499	87.9	84.2	73.9	81.1	95.7	92.4	92.3
BTS	0.8438	87.6	78.5	81.5	86.3	95.4	90.4	91.2
Multi-scale patch-based methods								
MOCNN	0.8642	89.2	80.1	85.8	87.3	96.0	93.3	92.0
PT-GID	0.8532	88.4	77.9	84.1	87.6	95.5	93.6	93.0
RMV	0.8619	89.1	79.6	87.9	85.2	95.3	93.9	93.1
OSA-CNN	0.8590	88.8	79.4	84.6	84.2	96.7	94.6	91.4
Position- and scale-aware method								
Proposed	0.9011	92.1	92.2	96.9	83.1	96.0	89.2	92.2

D. Experiment 3: ISPRS Vaihingen Dataset

The ISPRS Vaihingen dataset is a widely used benchmark for land-cover classification. It comprises 33 high-resolution orthoimages with a ground sampling distance of 9 cm, and provides annotations for six land-cover classes: Buildings, trees, cars, impervious surfaces, low vegetation, and clutter/background. To adapt the dataset for deep object classification, we randomly sample multiple points within the central region of each annotated object (connected component) and extract multiscale image patches centered on each point; the land-cover label of the sampled point is assigned to all corresponding patches, yielding a multiscale, center-aligned object-classification subset. In addition, the central 8×8-pixel label window of each patch is cropped to supervise semantic segmentation training. For evaluation, we designate image regions $\{10, 13, 17, 22, 32, 37\}$ as the test set, with the remaining regions used for training and validation.

1) Qualitative Comparison: Fig. 10 illustrates the classification results on the ISPRS Vaihingen dataset. Fixed-scale methods frequently break thin impervious structures (e.g., narrow streets) and leak across roof boundaries, and they often confuse low vegetation versus tree under shadows due to their inability to adapt patch scales and centers. MD multiscale baselines mitigate some of these issues yet remain constrained by predefined rules, leading to fragmented roads, perforated roofs, and the absorption of small cars into surrounding classes. In contrast, ObjectAgents adaptively selects both patch scales and positions, capturing key geometric cues (rectilinear roofs, junction topology) and contextual dependencies. As a result, it yields crisper building footprints, more continuous road networks, cleaner separation between low vegetation and trees, and better retention of small objects (cars), delivering superior visual fidelity especially in dense, structurally complex urban blocks.

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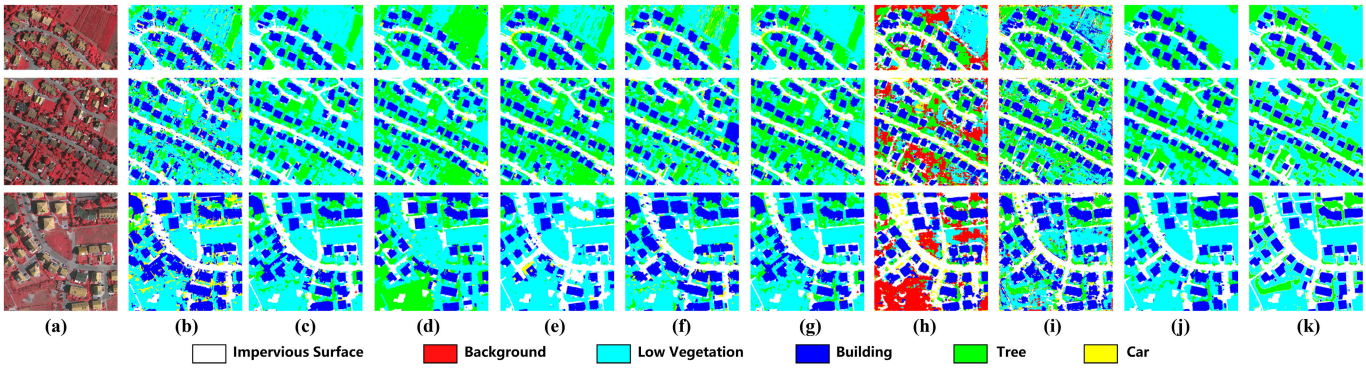


Fig. 10. Visual comparison of the classification results on the ISPRS Vaihingen dataset. (a) Image. (b) OCNN64. (c) BTS result. (d) RMV result. (e) PT-GID. (f) OSA result. (g) Multi OCNN. (h) ConvNeXt. (i) Segformer. (j) ObjectAgents. (k) Label.

TABLE III
COMPARISON OF DIFFERENT LAND-COVER CLASSIFICATION METHODS ON THE ISPRS VAIHINGEN DATASET

Method	Vaihingen dataset							
	Kappa	OA(%)	Impervious surfaces	Clutter/Background	Car	Tree	Low vegetation	Building
Fixed-scale patch-based methods								
Patch-based	0.5013	60.75	50.43	51.40	35.96	40.41	64.09	87.09
Fixed-scale	0.5505	65.60	69.18	29.41	30.67	60.48	88.17	50.87
ConvNeXt-UPerNet	0.6039	68.77	82.36	85.90	89.83	67.78	20.22	91.87
SegFormer	0.7216	78.76	87.10	69.03	79.81	82.61	49.56	89.69
BTS	0.5757	67.15	42.68	78.81	76.78	76.60	67.40	83.45
Multi-scale patch-based methods								
MOCNN	0.7016	77.34	62.69	78.16	60.70	84.96	75.61	87.84
PT-GID	0.6740	75.06	59.41	83.86	72.59	84.90	70.33	86.32
RMV	0.6047	69.80	43.96	66.94	63.18	69.04	76.30	87.84
OSA-CNN	0.6600	74.28	63.75	60.89	25.35	66.60	84.35	85.83
Position- and scale-aware method								
Proposed	0.7976	84.72	81.83	79.11	55.04	85.29	85.52	87.79

2) Quantitative Comparison: Table III summarizes the quantitative results on the ISPRS Vaihingen dataset. ObjectAgents attains the best overall performance with OA = 84.72% and $\kappa = 0.7976$, surpassing the strongest fixed-scale baseline SegFormer (78.76% OA, $\kappa = 0.7216$) by +5.96 pp in OA and +0.076 in κ, and exceeding the best multiscale baseline MOCNN (77.34% OA, $\kappa = 0.7016$) by +7.38 pp in OA and +0.096 in κ. Per-class analysis shows that ObjectAgents delivers the top scores on Tree (85.29%) and Low vegetation (85.52%), while remaining competitive on Building (87.79%) and Impervious surfaces (81.83%). We also note complementary strengths of other architectures: ConvNeXt-UperNet achieves the highest Car (89.83%), Background (85.90%) accuracies, and best Building (91.87%) result. These quantitative gains are consistent with the qualitative evidence in Fig. 10.

VI. ANALYSIS OF THE OBJECTAGENTS METHOD

A. Ablation Experiments

To rigorously validate the effectiveness of the proposed ObjectAgents framework and isolate the individual contributions of its core modules (i.e., the SCA and the PositionFilterAgent), we conducted a comprehensive ablation study. The experimental setup remained consistent with Section V-A, using ResNet-50 as the backbone.

1) Impact of the ScaleControlAgent (SCA): To evaluate the scale selection policy, we isolated the SCA by benchmarking it against alternative scale strategies while keeping the PFA constant. The comparative strategies are defined as follows:

1) *Fixed Scale and PFA:* Uses a traditional object-based CNN with fixed-size patches (128×128) to establish a baseline without scale adaptation.
2) *Random Scale and PFA:* Randomly executes scale actions (averaged over five independent runs to report mean and variance). This validates that the SCA's performance stems from learned policy optimization rather than random variation or data augmentation effects.
3) *MD Scale and PFA:* Sets the scale based on heuristic rules considering the distance-to-boundary. This tests the superiority of the learned DRL policy over rigid manual priors.
4) *Feature Ablation (PE Only/Context Only):* "PE only" trains the agent using only positional encoding (distance

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TABLE IV
ABLATION EXPERIMENT ON SCA

Method	OA	Kappa
Fixed scale + PFA	0.788	0.7642
Random scale + PFA	0.851	0.8343
MD scale + PFA	0.849	0.8321
PE only + PFA	0.845	0.8276
Context only + PFA	0.881	0.8676
ObjectAgents	0.907	0.8965

TABLE V
ABLATION EXPERIMENT ON THE PFA

Method	OA	Kappa
SCA + random position	0.819	0.7986
SCA + uniform position	0.857	0.8410
SCA + skeleton lines	0.880	0.8665
ObjectAgents	0.907	0.8965

metrics), while "Context only" uses only visual semantic features. This dissects the contribution of geometric versus semantic information in the decision-making process.

The results in Table IV demonstrate that the proposed ObjectAgents method (OA = 90.7%) significantly outperforms all baselines. Notably, the "Context only" strategy (OA = 88.1%) surpasses the "PE only" strategy (OA = 84.5%), indicating that semantic context is the primary driver for scale selection, though the combination of both (ObjectAgents) yields the optimal performance.

2) Impact of the PositionFilterAgent (PFA): To isolate the contribution of the boundary-biased sampling policy, we benchmarked the PFA against alternative positioning strategies while keeping the SCA constant:

1) *SCA and Random Position:* This strategy uses the SCA for scale selection and randomly selects positions within the target objects. This approach tests the robustness of the SCA when the position selection is random, helping to identify whether intelligent positional selection contributes significantly to performance.
2) *SCA and Uniform Position:* This strategy uses the SCA for scale selection with uniformly distributed positions across the object. This method evaluates the effect of a uniform position distribution on the overall classification accuracy, providing insights into the importance of positional uniformity.
3) *SCA and Skeleton Lines:* This strategy uses the SCA for scale selection and uses skeleton lines for position filtering. Skeleton lines are derived from the object's geometry and are used to guide the position selection process, aiming to capture the most structurally significant features within the object.

The performance metrics in Table V demonstrate that while geometric heuristics (Skeleton Lines) are competitive (OA = 88.0%), the proposed PFA achieves the highest accuracy (OA = 90.7%). This validates that the PFA module's ability to focus on ambiguous boundary regions is crucial for fine-grained classification.

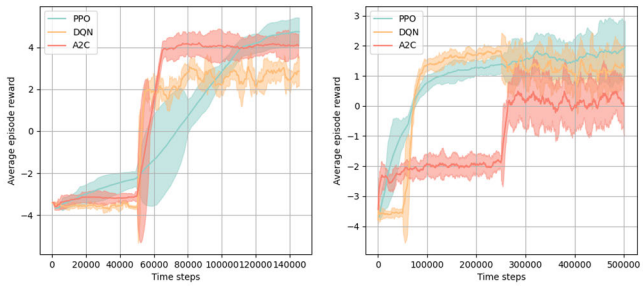


Fig. 11. Average performance of the various DRL methods on different datasets. (a) Global tidal wetlands. (b) IowaNet.

B. Sensitivity and Statistical Analysis

1) Statistical Significance and Stability: To address the necessity of statistical validation and ensure that our performance gains are not artifacts of random initialization, we analyzed the training stability and convergence variance. Fig. 11 reports the mean (solid lines) and standard deviation (shaded regions) of the average episode rewards across multiple independent runs on both the IowaNet and Global Tidal Wetlands datasets. In addition to the actor-critic (A2C) algorithm [64], the PPO algorithm was also employed for the comparative analysis. The deep Q-network (DQN) algorithm [44] was included as an additional baseline.

The experiments were designed to assess the performance and convergence behavior of different DRL algorithms in learning scale and position selection strategies. The models were trained on two datasets: IowaNet and the Global Tidal Wetlands dataset which is originally designed for object classification. The IowaNet dataset, with its substantial data volume, provides a rigorous test bed for evaluating the generalizability of DRL algorithms.

Convergence Dynamics: As illustrated in Fig. 11(a) and (b), while benchmarks like A2C and DQN eventually converge, the PPO demonstrates superior convergence efficiency. Specifically, PPO achieves a higher asymptotic reward with a steeper initial learning curve, indicating a more effective policy optimization process.

Variance and Confidence: The shaded regions represent the standard deviation, serving as a visual indicator of the confidence intervals. It is notable that the proposed method maintains a narrow error band throughout the training process, particularly in the later stages (after 100 000 time steps). This indicates low variance and high statistical robustness, confirming that the learned policies for scale and position selection are consistent across different random seeds.

2) Algorithmic Sensitivity: The comparison also highlights the sensitivity of the ObjectAgents framework to the underlying DRL algorithm. The instability of DQN (larger fluctuations in early stages) and the lower final reward of

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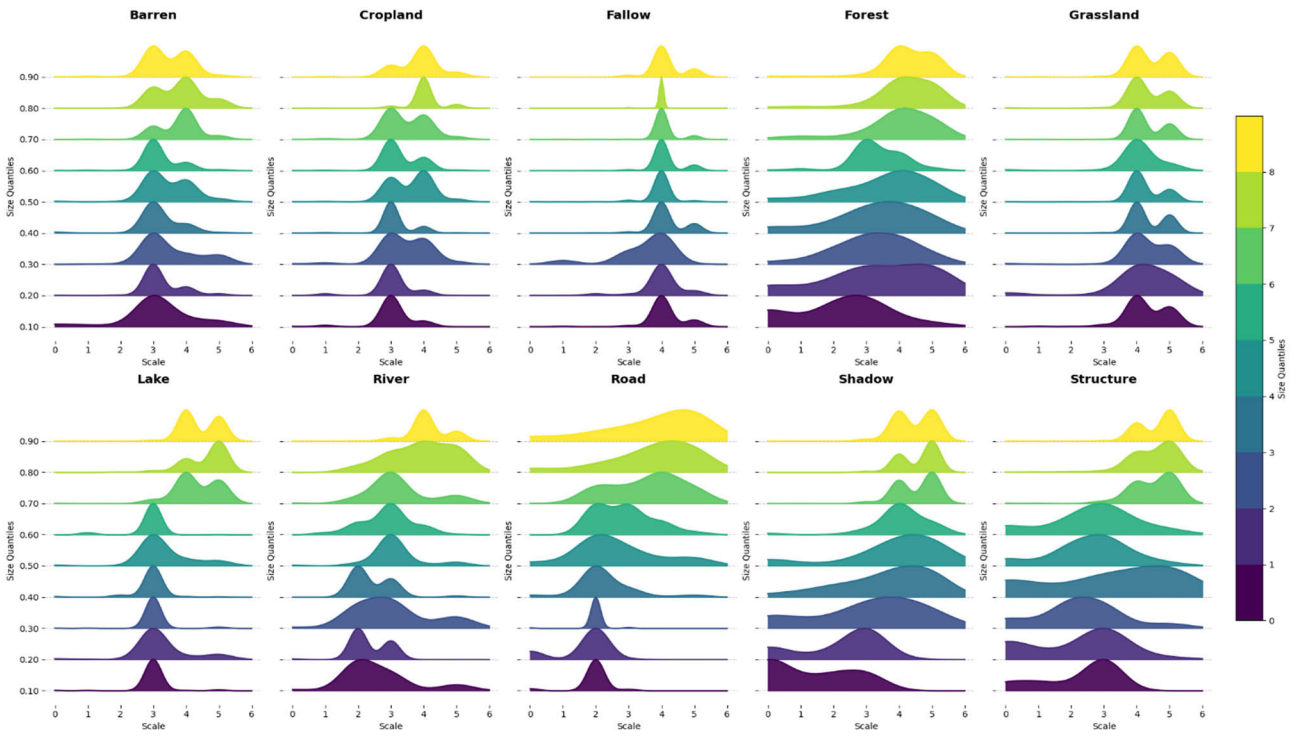


Fig. 12. Statistical analysis of the scales chosen for the different categories.

A2C suggest that the PPO algorithm's clipping mechanism is crucial for handling the continuous-discrete hybrid action space in our task.

These statistical analyses, combined with the component isolation in Section VI-A, confirm that the reported performance improvements are both statistically significant and directly attributable to the specific architectural designs of the SCA and PFA modules, rather than stochastic chance.

C. Visualization Analysis

We conducted a statistical analysis of the scales chosen for the different land-cover categories by dividing each object's area into 10 quantile levels, as shown in Fig. 12. For each quantile, we calculated the average scale of the image patches and examined the relationship between the selected scales and object sizes.

Consistent Scale Selection: For barren, cropland, fallow, and grassland areas, the scale selection remains consistent across the different object sizes, indicating a fixed-scale preference.

Adaptive Scale Selection: For lakes and rivers, the scale selection varies significantly with the object size, suggesting dynamic adjustment based on size and context.

Mixed Patterns: For forests, shadows, structures, and roads, the scale selection shows both fixed and adaptive patterns. Roads, similar to lakes and rivers, exhibit varied scale selection to capture both fine details and broader context.

The statistical analysis reveals that ObjectAgents exhibits distinct scale selection patterns based on the object category and size. For homogeneous land covers such as barren, cropland, fallow, and grassland, the scale remains relatively fixed. In contrast, for heterogeneous environments such as lakes, rivers, and roads, the agent adapts the scale based on the object size, highlighting the importance of contextual information and detailed feature capture.

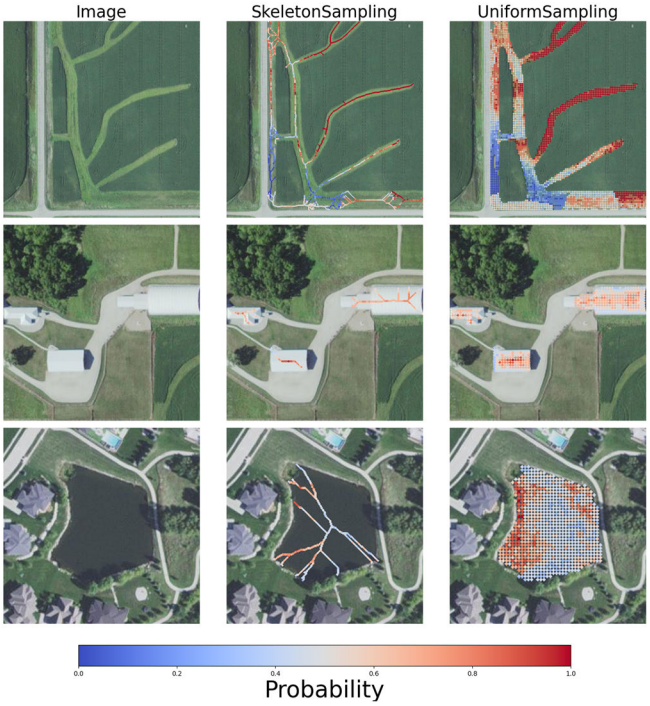


Fig. 13. Visualization of the results of the PositionFilterAgent.

We conducted a visualization analysis of agent actions by generating action maps. The agents executed actions at each point within the objects, including decisions on point

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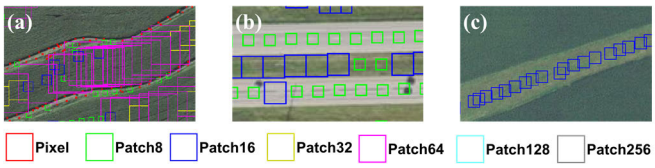


Fig. 14. Visualization of the results of the ScaleControlAgent. (a) Selected patch scale varies with object size; (b) even for objects with similar shapes and sizes, the agent makes different choices based on the object category; and (c) in grassland areas, smaller-scale patches yield better classification results for roads than geometrically matched scales.

elimination and scale selection for candidate points. The maps shown in Fig. 13 illustrate the actions taken by agents across different positions and land-cover categories, with the action probability maps generated using the PPO algorithm. Candidate points for the PFA were generated using uniform sampling, while skeleton sampling was used for the SCA. To verify the effectiveness of the proposed feature indexing method, we compared the action maps of the network trained with the original CNN model.

Fig. 13 illustrates the action probability maps for the PFA using the skeleton sampling and uniform sampling methods. The images on the left provide the original high-resolution remote sensing context. The color bar at the bottom indicates the probability scale, with blue representing low probability (0.0) and red representing high probability (1.0).

Agricultural Fields (First Row): The agent reduces the selection probability in areas with blurred boundaries near the grassland and road intersections. This adaptive behavior helps avoid selecting image patches with lower discriminative power, unlike the rule-based strategies that fail to adapt, and affect the classification performance.

Industrial Area (Second Row): Both the skeleton sampling and the agent's strategy focus on centrally located candidate points within the objects. For buildings, selecting homogeneous image patches is crucial for accurate classification.

Lakes (Third Row): The agent's strategy focuses on edge points, emphasizing the importance of boundary information due to the richer contextual data available, contrasting with skeleton sampling, which focuses on central regions.

These maps provide significant insights into the agent's adaptive decision-making process for selecting candidate points, demonstrating an effective approach to handling diverse land-cover complexities.

The visualization results of the SCA (Fig. 14) reveal that the scale of the selected image patches varies with the object size [Fig. 14(a)]. Even for objects with similar shapes and sizes, the agent makes different choices based on the object category [Fig. 14(b)].

For grassland objects, smaller-scale patches are preferred, and the classification results show that smaller scales yield better results for roads in grassland areas, compared to geometrically matched sizes [Fig. 14(c)]. These findings illustrate that the SCA adjusts the scale of the candidate point image patches based on the contextual information needs.

D. Discussion and Limitations

Unlike traditional CNNs or rule-based methods, ObjectAgents leverage reinforcement learning to enable dynamic, object-centric decision-making. Through the collaboration of the PFA and SCA, the framework lets the agents explore the environment and adaptively select optimal spatial scales, rather than relying on fixed, manually defined strategies. This flexibility enhances classification, especially for challenging categories like rivers, fallow land, and aquaculture, which are often misclassified due to boundary ambiguities.

Attention and transformer decoders aggregate multiscale context via soft weights, but they do not directly optimize the nondifferentiable, budgeted decisions central to object-centric classification: 1) retain versus discard candidate points and 2) choose one discrete patch scale. ObjectAgents tackles these with *RL* and task-aligned rewards (cross-scale consistency; boundary-aware regularization), yielding interpretable policies.

In our benchmarks, transformer/global-context baselines (SegFormer, ConvNeXt-UPerNet) remain below ObjectAgents (e.g., 90.7% versus 87.0% OA on IowaNet; 92.1% versus 87.9% on Wetlands; 84.72% versus 78.76% on Vaihingen). Moreover, our ablation "Context-only + PFA" (Table IV)—which approximates attention-like aggregation across scales—still underperforms ObjectAgents by 2.6% OA, indicating that explicit discrete scale policies add value beyond soft aggregation.

Beyond superior classification accuracy, computational efficiency and scalability are critical for practical deployment. Although DRL training is iterative, the proposed framework is optimized for efficiency. We employ a "frozen environment" strategy to avoid costly backbone updates during *RL* optimization. ObjectAgents functions as a dynamic pruning mechanism. By filtering noninformative regions (PFA) and selecting a single optimal scale (SCA), it eliminates the redundancy of dense multiscale pyramids. This significantly reduces FLOPs per object, ensuring scalability for large-scale datasets.

Compared with SOTA region-adaptive networks and deformable convolutions, ObjectAgents offers a more interpretable, object-centric sampling mechanism that aligns well with remote-sensing workflows. That said, the approach has limitations: training can be sensitive to compute budget and hyperparameters—especially the reward design, which embeds task priors. Although inference is efficient, broader evaluation is needed to support large-scale deployment, where generalization may be stressed. In addition, the *RL* backbones used here (e.g., PPO, A2C) are generic; injecting remote-sensing–specific inductive biases (e.g., geospatial constraints, physical scales) may further improve robustness. As future work, we will explore integrating geospatial knowledge to better guide the agents during training.

VII. Conclusion

In this article, we introduced ObjectAgents to overcome static sampling limitations in OBIA. Unlike prior DRL approaches, it uniquely decouples sampling into cooperative

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agents (PFA and SCA), enabling interpretable, sequential optimization of position and scale. By modeling the classification task as an MDP, the PFA and SCA dynamically determine the optimal position and scale for image patches. In addition, an object-based multiscale classification network is used to accurately label remote sensing objects. These two networks are alternately trained using supervised learning and DRL. The experimental results on three datasets demonstrated that ObjectAgents outperforms the current multiscale object-based deep learning classification methods. Sensitivity analyses confirmed the robustness of ObjectAgents, and visualization analyses highlighted its precision in position and scale selection, indicating strong potential for broader deployment; nevertheless, verifying robustness across additional regions, sensors, and taxonomies remains future work.

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